

POINTE DE PENMARCH, France

WMO: 072000

Lat: 47.7976N

Lon: 4.3747W

Elev: 19

StdP: 101.10

Time Zone: 1.00 (EUC)

Period: 94-14

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	0.1	1.5	-4.7	2.5	1.7	-2.9	3.0	2.9	21.8	9.6	20.2	9.7	4.2	60	0.707

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.0	23.8	18.7	22.2	18.2	21.0	17.7	19.9	22.5	19.1	21.2	18.4	20.2	4.2	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.0	13.8	21.0	18.2	13.1	20.1	17.5	12.6	19.5	57.0	22.8	54.2	21.3	52.0	20.3	24.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.3	16.1	14.3	-1.9	27.5	2.4	2.4	-3.6	29.3	-5.0	30.7	-6.3	32.1	-8.1	33.8		
(5)			-3.2	21.1	1.5	1.6	-4.3	22.2	-5.2	23.1	-6.0	24.0	-7.1	25.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	12.8	8.4	8.4	9.5	11.0	13.6	16.0	17.4	17.7	16.5	14.5	11.6	9.0		
	DBStd	4.14	3.12	2.47	2.12	2.10	2.16	2.01	1.98	1.60	1.85	2.07	2.35	3.22		
	HDD10.0	232	63	53	33	13	1	0	0	0	0	1	12	56		
	HDD18.3	2047	308	279	274	220	149	74	43	32	60	118	202	288		
	CDD10.0	1264	13	7	17	43	111	181	229	238	195	142	60	26		
	CDD18.3	37	0	0	0	0	1	6	14	12	5	0	0	0		
	CDH23.3	82	0	0	0	0	3	13	38	18	10	0	0	0		
	CDH26.7	12	0	0	0	0	0	2	6	2	2	0	0	0		
(14) Wind	WSAvg	6.8	7.9	7.6	6.7	6.8	6.5	5.9	5.8	5.7	5.7	7.0	8.2	8.5		
(15) Precipitation	PrecAvg	861	102	84	59	67	59	44	48	50	55	89	104	98		
	PrecMax	1105	222	196	186	166	147	97	134	102	150	135	214	186		
	PrecMin	628	23	17	7	3	5	5	10	5	3	21	19	20		
	PrecStd	115	52	45	39	40	31	25	28	28	37	30	49	44		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.2	13.1	16.9	18.4	23.0	25.1	27.4	25.7	24.4	19.8	16.8	14.6		
		MCWB	12.1	11.2	11.7	13.7	16.4	18.9	19.8	20.7	18.8	17.0	15.5	13.2		
	2%	DB	12.6	12.2	14.1	16.8	20.5	22.4	23.8	22.9	22.0	18.4	15.7	14.0		
		MCWB	11.7	11.0	10.9	12.9	15.6	17.7	19.0	19.3	17.6	16.6	14.1	12.8		
	5%	DB	12.2	11.8	12.7	15.4	18.5	20.8	21.8	21.4	20.6	17.8	15.2	13.5		
		MCWB	11.2	10.7	10.6	12.3	14.9	16.9	18.5	18.7	17.4	16.2	13.8	12.3		
	10%	DB	11.7	11.3	12.0	14.0	16.9	19.3	20.4	20.2	19.5	17.1	14.8	13.0		
		MCWB	10.7	10.2	10.4	11.5	14.0	16.4	17.7	18.0	16.9	15.7	13.4	11.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.6	12.1	12.6	14.9	17.5	20.0	20.7	21.2	20.1	18.0	15.6	13.7		
		MCDB	12.8	12.4	15.2	17.4	21.2	23.5	24.3	23.9	22.9	18.9	16.2	14.1		
	2%	WB	12.1	11.6	11.8	13.7	16.2	18.5	19.6	20.1	18.9	17.2	14.8	13.1		
		MCDB	12.5	12.0	12.9	15.9	19.3	21.3	22.8	22.0	20.7	18.0	15.3	13.7		
	5%	WB	11.5	11.1	11.4	12.7	15.3	17.6	18.7	19.2	18.1	16.6	14.3	12.6		
		MCDB	12.0	11.6	12.3	14.4	17.9	20.0	21.1	20.9	19.7	17.3	15.1	13.3		
	10%	WB	11.0	10.5	10.9	12.1	14.4	16.7	18.0	18.4	17.4	16.0	13.6	12.1		
		MCDB	11.6	11.2	11.7	13.3	16.4	18.8	20.0	19.9	19.0	16.8	14.6	12.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	3.2	3.7	4.1	4.6	4.4	4.4	4.0	4.0	4.7	3.5	3.4	3.6		
		MCDBR	2.2	3.1	5.3	7.1	7.7	7.8	7.2	6.4	7.1	3.8	3.3	2.5		
	5% WB	MCWBR	2.6	3.0	3.9	4.7	4.9	5.0	4.3	4.0	4.4	3.0	3.2	3.1		
		MCWBR	2.1	2.4	3.8	5.1	6.5	6.5	5.8	5.2	5.3	3.0	2.6	2.4		
(40) Clear-Sky Solar Irradiance	taub		0.340	0.352	0.377	0.386	0.389	0.386	0.384	0.374	0.362	0.362	0.342	0.333		
		taud	2.400	2.389	2.342	2.338	2.375	2.412	2.419	2.466	2.478	2.460	2.427	2.373		
	Ebn at Noon		725	801	837	866	875	878	873	869	846	782	716	679		
		Edh at Noon	70	87	105	116	116	113	111	101	91	80	68	64		
(44) All-Sky Solar Radiation	RadAvg		1.02	1.88	3.11	4.82	5.67	6.14	5.88	5.07	4.05	2.37	1.29	0.86		
	RadStd		0.08	0.17	0.33	0.39	0.46	0.63	0.28	0.39	0.29	0.18	0.09	0.09		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (20 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page